

Study Guide 2: Bootstrapping the Spot Curve

Fixed Income Study Guides

Bootstrapping extracts zero-coupon (spot) rates from coupon bond prices, one maturity at a time. Each step prices the earlier cash flows with rates already found, leaving a single unknown to solve.

1. The idea

- A coupon bond is a bundle of zeros. Its price equals the sum of each cash flow times the discount factor for its date.
- The shortest bond has one cash flow, so it gives the first rate directly.
- Each longer bond adds exactly one new date. Subtract the value of its known earlier cash flows and solve for the new date's discount factor.

2. Core formulas

Price as a sum of discounted cash flows, with DF_k the discount factor for period k :

$$P_n = \frac{c}{2} \sum_{k=1}^n DF_k + 100 DF_n$$

Solved for the new discount factor (the bootstrap step):

$$DF_n = \frac{P_n - \frac{c}{2} \sum_{k=1}^{n-1} DF_k}{100 + \frac{c}{2}}$$

Spot rate from a discount factor (semiannual compounding):

$$s_n = 2 \left[DF_n^{-1/n} - 1 \right]$$

Forward rate for the period from $n - 1$ to n :

$$f_{n-1,n} = 2 \left[\frac{DF_{n-1}}{DF_n} - 1 \right]$$

Working in discount factors is simpler than working in spot rates: every step is linear until the final conversion.

3. Example A: annual coupons

Three par bonds (price 100), annual coupons, used only to show the logic.

Maturity	Coupon	Known cash flows valued at	Remaining value	Spot rate
1 yr	3%	none	100	3.00%
2 yr	4%	$4 \div 1.03 = 3.883$	96.117	4.02%
3 yr	5%	$4.854 + 4.621 = 9.475$	90.525	5.07%

The 3-year step:

$$(1 + s_3)^3 = \frac{105}{90.525} = 1.1599 \Rightarrow s_3 = 5.07\%$$

4. Example B: semiannual coupons (Treasury convention)

Four par bonds, coupons paid twice a year, rates annualized with semiannual compounding.

Maturity	Coupon (annual)	Paid per period	Earlier coupons valued at	Discount factor	Spot rate	Forward rate for that period
0.5 yr	4.0%	2.00	none	0.980392	4.0000%	4.0000%
1.0 yr	4.2%	2.10	2.0588	0.959267	4.2021%	4.4044%
1.5 yr	4.4%	2.20	4.2673	0.936719	4.4059%	4.8143%
2.0 yr	4.6%	2.30	6.6157	0.912848	4.6117%	5.2302%

Step for 2.0 years: remaining value = $100 - 6.6157 = 93.3843$. Then

$$\left(1 + \frac{s}{2}\right)^4 = \frac{102.30}{93.3843} = 1.095473 \Rightarrow s = 4.6117\%$$

Check: the 2-year discount factors reprice the 4.6% bond at exactly 100.

5. Par, spot and forward curves

- **Par curve:** the coupon at which a bond of each maturity prices at 100. This is what most people mean by "the yield curve."
- **Spot curve:** the zero-coupon rate for each date. A par yield is a weighted blend of spot rates.
- **Forward curve:** the rate implied for each future period. Spot rates are geometric averages of forwards.
- Upward-sloping curve: forward > spot > par at longer maturities, as in Example B (5.23% > 4.61% > 4.60% at 2 years). Inverted curve: the order flips. Flat curve: all three coincide.

6. In practice

- **Inputs:** Treasury bills at the short end, then notes and bonds. Many desks prefer off-the-run issues, because on-the-run issues carry a liquidity premium.
- **Gaps:** bonds rarely mature on every date needed. Fill them by interpolation (linear on spot rates, or log-linear on discount factors), or fit a smooth curve.
- **Fitted models:** Nelson-Siegel and Svensson describe the whole curve with 4-6 parameters. They are smoother but no longer reprice every bond exactly.
- **Day counts:** real Treasury periods follow Actual/Actual, so periods are not exactly 0.5 years. Settlement between coupon dates requires accrued interest and fractional periods.
- **Error propagation:** a bad short-end price flows into every longer rate.
- **Other inputs:** swap and SOFR curves are bootstrapped the same way from deposits, futures and swaps.

7. Review questions

1. Why does the bootstrap start at the shortest maturity? *Its bond has a single cash flow, so its rate is known directly.*
2. Using Example B's discount factors, price a 1.5-year 5% bond. *Answer: $2.5 \times (0.980392 + 0.959267 + 0.936719) + 100 \times 0.936719 = 7.192 + 93.672 = 100.86$.*
3. From Example B, what is the forward rate from 1.0 to 1.5 years? *Answer: $2 \times (0.959267/0.936719 - 1) = 4.81\%$.*
4. When would the spot curve lie below the par curve? *When the curve is inverted.*
5. Why does a fitted Nelson-Siegel curve not reprice every bond exactly? *It trades exact fit for smoothness, which filters out noise from individual bonds.*

8. CFA-style review questions

Original questions in the CFA Level I format (three choices, one correct). Not CFA Institute material. Assume annual compounding.

1. The 1-year spot rate is 3.00%. A 2-year, 4% annual-coupon bond trades at par. The 2-year spot rate is closest to:
 - A. 4.00%
 - B. 4.02%
 - C. 4.04%
2. The 1-year spot rate is 3% and the 2-year spot rate is 4%. The 1-year forward rate one year from now is closest to:
 - A. 4.50%
 - B. 5.01%
 - C. 6.00%
3. Using spot rates of 3% (1 year) and 4% (2 years), the price of a 2-year 5% annual-coupon bond is closest to:
 - A. 101.93
 - B. 101.89
 - C. 102.00
4. When the par yield curve slopes upward, the spot curve at longer maturities is:
 - A. below the par curve
 - B. above the par curve
 - C. identical to the par curve
5. Valuing a bond with spot rates rather than its own yield to maturity is preferred because:
 - A. it gives an arbitrage-free value consistent with stripping and reconstitution
 - B. spot rates are always lower than yields to maturity
 - C. it removes interest rate risk from the valuation

Answer key

1. B. $100 = 4/1.03 + 104/(1 + s_2)^2$, so $s_2 = 4.02\%$.
2. B. $1.04^2/1.03 - 1 = 5.01\%$.
3. A. $5/1.03 + 105/1.04^2 = 4.854 + 97.078 = 101.93$.
4. B. A par yield blends lower short-term spot rates, so it sits below the long spot rate.
5. A. Each cash flow is priced at its own date's rate, so no stripping profit is possible.

9. Bloomberg Terminal exercises (draft: verify on the campus Terminal)

Command	What to do	Check against this guide	Verified
CRVF <G0>	Curve finder: locate the U.S. Treasury active curve and its zero (spot) version	Section 5: par vs spot curves	☐
GC <G0>	Plot par, zero and forward versions of the Treasury curve on one chart	Upward-sloping curve: forward > spot > par	☐
ICVS <G0>	Curve builder: view the instruments behind the curve and its zero rates and discount factors	Sections 2 and 6: inputs, interpolation	☐
FWCV <G0>	Forward curve analysis	Section 2: forward rate formula	☐
BDS in Excel	Pull the curve's members, e.g. <code>=BDS("YCGT0025 In-dex", "CURVE_TENOR_RATES")</code> (confirm the curve ticker in CRVF)	Inputs for the Excel bootstrap in Study Guide 3	☐

Exercise: take the 6-month, 1-year, 1.5-year and 2-year Treasury yields from the Terminal, bootstrap them with the Excel layout from Study Guide 3, and compare your spot rates with the Terminal's zero curve. Differences should be small; explain them (interpolation method, day counts, bills vs coupon securities at the short end).

10. Reading in Bodie, Kane & Marcus, *Investments*, 13th ed. (2024)

Section	Topic in the book	Supports
15.1 The Yield Curve	Bond Pricing	Spot rates and pricing coupon bonds off the zero curve; the foundation for bootstrapping
15.2 The Yield Curve and Future Interest Rates	The Yield Curve under Certainty; Holding-Period Returns; Forward Rates	Deriving forward rates from spot rates
15.3 Interest Rate Uncertainty and Forward Rates		Forwards vs expected future rates; term premium
15.4 Theories of the Term Structure	Expectations Hypothesis; Liquidity Preference; Market Segmentation	Why the curve slopes as it does
15.5 Interpreting the Term Structure		Reading the curve's shape
15.6 Forward Rates as Forward Contracts		Locking in forward rates with zeros
14.2 Bond Pricing	Bond Pricing between Coupon Dates	Fractional periods and accrued interest

The book builds the zero curve and forwards but gives the step-by-step bootstrap only lightly;

this guide's worked examples fill that gap.